PALAK PANDIT

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ACADEMIC PROFILE

Bachelor of Engineering in Information Science and Engineering
Nitte Meenakshi Institute of Technology, Bengaluru
CGPA: 9.25
(Aug. 2023 - Jul. 2027)

SKILLS

Programming Languages: Python, C++, Java, SQL
Machine Learning: Supervised Learning, Neural Networks, CNNs, Transformers, Classification, Regularization, Gradient-Based Optimization
Generative AI : Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search
Agentic AI & Document Intelligence: Agentic AI, Information Extraction, Structured Outputs, LLM-Based Reasoning
Evaluation & Experimentation: Model Evaluation, Precision, Recall, Error Analysis, Data Validation, Exploratory Data Analysis (EDA)
AI/ML Tools & Libraries: PyTorch, scikit-learn, Pandas, NumPy, FAISS, SentenceTransformers, Google Gemini API, Matplotlib
Frameworks & Development Tools: Django, Streamlit, Git, GitHub, BeautifulSoup, SQLite, MySQL

PROJECTS

E-Store – Modular E-Commerce Web Application May 2024 – July 2024
- **Tech Stack:** Python, Django, SQLite, HTML5, CSS, JavaScript.
- Developed a modular, full-featured e-commerce platform using **Django** with reusable apps for authentication, catalog and cart.
- Implemented **secure user management** — registration, authentication (login/logout) and profile updates
- Built **category-based browsing**, **search**,plus a **shopping cart & checkout** supporting add/remove/update and order placement.

ChessBot – Interactive Chess Engine with AI Opponent Nov 2025 – Dec 2025
- **Tech Stack:** Python, Pygame, Minimax, Alpha-Beta Pruning, Heuristics, and Object-Oriented Programming concepts.
- Built an interactive chess application with animated moves and legal move validation.
- Implemented undo/redo controls and complete game-state handling for checkmate, stalemate, repetition cases.
- Created an AI opponent using Minimax, Alpha-Beta pruning, and heuristic evaluation strategies.

AI Chatbot for GitLab Documentation Jan 2026 – Feb 2026
- **Tech Stack:** Python, Streamlit, FAISS, SentenceTransformers, Google Gemini, BeautifulSoup
- Engineered a Retrieval-Augmented Generation (RAG) chatbot for answering questions from GitLab handbook documentation.
- Built an data pipeline for web crawling, text preprocessing, document chunking, embedding generation, and FAISS indexing.
- Implemented semantic similarity search to retrieve relevant context and integrated Gemini for context-aware response generation.
- Added multi-turn conversation support, source citation, and session management through an interactive Streamlit interface.

CIFAR-10 Image Classification using CNN Mar 2026 – Apr 2026
- **Tech Stack:** Python, PyTorch, CNN, NumPy, Matplotlib, scikit-learn.
- Built a **PyTorch CNN** to classify CIFAR-10 images accurately across **10 distinct object categories**.
- Implemented **convolutional layers and ReLU activations** to learn hierarchical visual features from input images.
- Applied **max pooling and fully connected layers** for dimensionality reduction and final image classification.
- Trained the neural network using **cross-entropy loss** and **Adam optimizer** for efficient convergence.
- Evaluated classification performance using **accuracy, precision, recall, F1-score, and confusion matrix** across classes.
- Applied **dropout regularization** to control overfitting and improve generalization performance on unseen images.
- Monitored **training and validation losses** across epochs to analyze convergence and identify potential overfitting.

CERTIFICATIONS

Data Scientist Masters Program – Simplilearn **Apr 2026**
Advisory Launchpad Program by PwC **Jul 2026**

HONOURS AND ACHIEVEMENTS

- **Coding Profiles:** Codeforces – `palakpandit43`, Leetcode – `heypalak11`, Solved over **300+** LeetCode problems.
- **Code Cadets Hackathon** – Secured **top 5%** in IISc's hackathon among **500+ participants**.
- **GDG (Google Developer Group):** Led marketing and execution for **5+** technical events.
- Built a campus developer community and organized workshops with **200+** attendance.
- Achieved **96 percentile** in **JEE Mains 2023**, ranking among one million candidates.